

A smooth double-cone answer to Question 4.12 of Guerra–Raita–Schrecker

Run `fa_banach_001`, lane 2

2026-07-18

Status. This is a full solution packet for the exact smooth form of Question 4.12 from Guerra–Raita–Schrecker. It does not claim to solve the separate rough BV^{Div} problem discussed near that question.

Source. The source paper is Andre Guerra, Bogdan Raita, Matthew Schrecker, *Compensation phenomena for concentration effects via nonlinear elliptic estimates*, arXiv:2112.10657. Question 4.12 appears on page 28 of the source PDF:

no estimate for fields constrained to take values in $\text{Sym}_2 \cap \{\det \leq 0\}$. Note that $\mathcal{K}_2^- \cap \text{Sym}_2$ is a two-dimensional half-space, while $\tilde{A}$ takes values in the full linear space containing this half-space. Thus entrywise constraints are in some sense necessary for an estimate.

4.4. Density of smooth functions and separation. The previous two remarks provide examples of matrix fields A which take values in $\text{Sym}_2 \cap \{\det \leq 0\}$ and which falsify the inequalities of this section. Notice also that, in both cases, the matrix field A is *constant* on the boundary of the unit ball. A question which the previous remarks leave open is the following:

Question 4.12. *Can the inequalities of this section hold if we constrain the fields to take values in the double cone*

$$\text{Sym}_2 \cap \{\det \geq 0\} = \text{Sym}_2 \cap (\mathcal{K}_1^+ \cup \mathcal{K}_3^+) = \text{Sym}_2^+ \cup \text{Sym}_2^-?$$

In fact, for fields satisfying the exact constraint $\text{Curl } A = 0$, we have the following rigidity result, which is a particular case of a theorem of Šverák [69], see also the remarkable generalisations due to Faraco–Székelyhidi [27] as well as [71, 72, 73] for related results:

Proposition 4.13. *Let $A \in L^2(\mathbb{B}^2, \text{Sym}_2)$ be such that $\text{Curl } A = 0$ and $A|_{\mathbb{S}^1}$ is constant. Then*

$$\text{either } A \in \text{Sym}_2^+ \text{ a.e. in } \Omega \quad \text{or } A \in \text{Sym}_2^- \text{ a.e. in } \Omega.$$

Transcription of the target question. Question 4.12 asks whether the inequalities of the two-dimensional section can hold when the fields are constrained to the double cone

$$\text{Sym}_2 \cap \{\det \geq 0\} = \text{Sym}_2^+ \cup \text{Sym}_2^-.$$

The inequalities in that section include the row-divergence determinant estimate of Theorem 4.1, its L^2 consequence, and the row-curl determinant estimate of Corollary 4.7.

Notation. If $A = (a_{ij})$ is a 2×2 matrix field, let A_i be its i -th row and define

$$d_i(A) = \text{div } A_i = \partial_1 a_{i1} + \partial_2 a_{i2}, \quad r_i(A) = \text{curl } A_i = \partial_1 a_{i2} - \partial_2 a_{i1}.$$

We use the Frobenius matrix norm in the proof.

Proof Intuition

A smooth field valued in $\text{Sym}_2^+ \cup \text{Sym}_2^-$ can change branches only by passing through the zero matrix: if a matrix in the double cone has trace zero, then both eigenvalues vanish. Therefore the sign of $\text{tr } A$ can be used to flip the negative branch into the positive one without creating a hidden first-order term. To keep the proof smooth, use

$$B_\varepsilon = \frac{\text{tr } A}{\sqrt{(\text{tr } A)^2 + \varepsilon^2}} A.$$

Each B_ε is positive semidefinite, so the source paper's positive-cone estimates apply. The determinant converges to $\det A$, and the extra terms coming from differentiating the scalar factor vanish in L^1 because $|A| \leq |\text{tr } A|$ on the double cone.

Main Result

Theorem 1 (Smooth double-cone estimates). *Let $A \in C_c^\infty(\mathbb{R}^2, \text{Sym}_2^+ \cup \text{Sym}_2^-)$. Then*

$$\int_{\mathbb{R}^2} \det A \, dx \leq \|d_1(A)\|_{L^1(\mathbb{R}^2)} \|d_2(A)\|_{L^1(\mathbb{R}^2)}$$

and

$$\int_{\mathbb{R}^2} \det A \, dx \leq \|r_1(A)\|_{L^1(\mathbb{R}^2)} \|r_2(A)\|_{L^1(\mathbb{R}^2)}.$$

Proof. Write $\tau = \text{tr } A$, and for $\varepsilon > 0$ define

$$\eta_\varepsilon(t) = \frac{t}{\sqrt{t^2 + \varepsilon^2}}, \quad B_\varepsilon = \eta_\varepsilon(\tau)A.$$

Since $A(x)$ is either positive semidefinite or negative semidefinite, the scalar $\eta_\varepsilon(\tau(x))$ has the sign needed to make $B_\varepsilon(x) \in \text{Sym}_2^+$. Thus $B_\varepsilon \in C_c^\infty(\mathbb{R}^2, \text{Sym}_2^+)$. Also

$$\det B_\varepsilon = \eta_\varepsilon(\tau)^2 \det A \rightarrow \det A$$

pointwise and in L^1 , since A is smooth and compactly supported.

We record the elementary bound used to control the derivative of $\eta_\varepsilon(\tau)$. If $M \in \text{Sym}_2^+ \cup \text{Sym}_2^-$, then

$$|M|_F \leq |\text{tr } M|.$$

Indeed, on either branch the eigenvalues have the same sign, so $(\lambda_1^2 + \lambda_2^2)^{1/2} \leq |\lambda_1 + \lambda_2|$. In particular, $|A_{ij}| \leq |\tau|$.

For a row divergence we have

$$d_i(B_\varepsilon) = \eta_\varepsilon(\tau)d_i(A) + \eta'_\varepsilon(\tau)(A_{i1}\partial_1\tau + A_{i2}\partial_2\tau).$$

For a row curl,

$$r_i(B_\varepsilon) = \eta_\varepsilon(\tau)r_i(A) + \eta'_\varepsilon(\tau)(A_{i2}\partial_1\tau - A_{i1}\partial_2\tau).$$

Here

$$\eta'_\varepsilon(t) = \frac{\varepsilon^2}{(t^2 + \varepsilon^2)^{3/2}}, \quad |t\eta'_\varepsilon(t)| \leq 1, \quad t\eta'_\varepsilon(t) \rightarrow 0.$$

Using $|A_{ij}| \leq |\tau|$, the error terms above are dominated by an integrable multiple of $|\nabla \tau|$ and converge pointwise to zero. Dominated convergence gives

$$\limsup_{\varepsilon \downarrow 0} \|d_i(B_\varepsilon)\|_1 \leq \|d_i(A)\|_1, \quad \limsup_{\varepsilon \downarrow 0} \|r_i(B_\varepsilon)\|_1 \leq \|r_i(A)\|_1.$$

Since $\text{Sym}_2^+ \subset \{a_{11}, a_{22} \geq 0\}$, Theorem 4.1 of the source paper applies to each B_ε :

$$\int_{\mathbb{R}^2} \det B_\varepsilon dx \leq \|d_1(B_\varepsilon)\|_1 \|d_2(B_\varepsilon)\|_1.$$

Letting $\varepsilon \downarrow 0$ gives the first estimate. Likewise, Corollary 4.7 of the source paper gives

$$\int_{\mathbb{R}^2} \det B_\varepsilon dx \leq \|r_1(B_\varepsilon)\|_1 \|r_2(B_\varepsilon)\|_1,$$

and the same limit gives the row-curl estimate. □

Corollary 1 (Smooth L^2 consequences). *Let $\mathcal{C} \subset \text{Sym}_2^+ \cup \text{Sym}_2^-$ be a closed cone such that*

$$|M|^2 \leq K \det M \quad (M \in \mathcal{C})$$

for some $K < \infty$. Then every $A \in C_c^\infty(\mathbb{R}^2, \mathcal{C})$ satisfies

$$\|A\|_{L^2(\mathbb{R}^2)} \leq C_K \|\text{Div } A\|_{L^1(\mathbb{R}^2)}$$

and

$$\|A\|_{L^2(\mathbb{R}^2)} \leq C_K (\|r_1(A)\|_{L^1(\mathbb{R}^2)} + \|r_2(A)\|_{L^1(\mathbb{R}^2)}).$$

Proof. The cone assumption and the theorem give

$$\|A\|_2^2 \leq K \int_{\mathbb{R}^2} \det A \leq K \|d_1(A)\|_1 \|d_2(A)\|_1 \leq C_K \|\text{Div } A\|_1^2.$$

The curl estimate is identical, using $\|r_1(A)\|_1 \|r_2(A)\|_1 \leq (\|r_1(A)\|_1 + \|r_2(A)\|_1)^2/4$. □

Scope and Limitations

This resolves Question 4.12 for the smooth compactly supported estimates in Section 4. The paper separately discusses rough fields in BV^{Div} , where the usual convex-cone mollification argument fails. The theorem above does not solve that rough problem. For a rough field with a jump directly from a positive semidefinite value to a negative semidefinite value, a branch flip can introduce a new jump contribution in the divergence measure.

Novelty Check

Bounded checks on 2026-07-18 searched the run indexes, the local parsed arXiv corpus, and public web search for exact phrases including *Question 4.12*, *double cone*, $\text{Sym}_2^+ \cup \text{Sym}_2^-$, *Can the inequalities of this section hold*, and the source arXiv id. No separate later answer was found; exact hits led back to the source paper.

References

- [1] A. Guerra, B. Raita, and M. Schrecker, *Compensation phenomena for concentration effects via nonlinear elliptic estimates*, Ars Inveniendi Analytica (2024), Paper No. 1; arXiv:2112.10657.